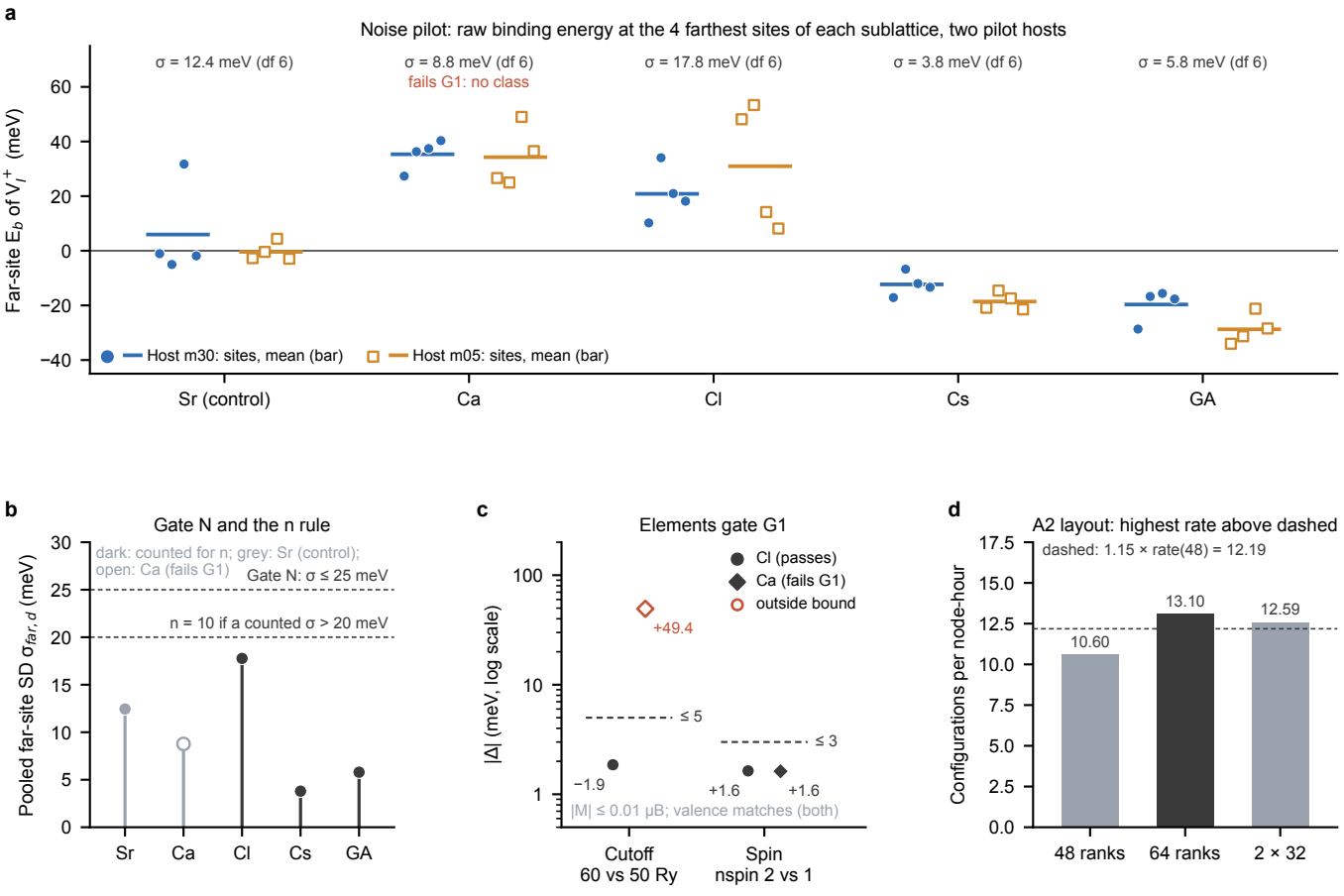

a
Noise pilot: raw binding energy at the 4 farthest sites of each sublattice, two pilot hosts
σ = 12.4 meV (df 6)
σ = 8.8 meV (df 6)
fails G1: no class
σ = 17.8 meV (df 6)
σ = 3.8 meV (df 6)
σ = 5.8 meV (df 6)
Far-site E_b of V_I^+ (meV)
60
40
20
0
−20
−40
Host m30: sites, mean (bar)
Host m05: sites, mean (bar)
Sr (control)
Ca
Cl
Cs
GA
b
Gate N and the n rule
dark: counted for n; grey: Sr (control);
open: Ca (fails G1)
Gate N: σ ≤ 25 meV
n = 10 if a counted σ > 20 meV
Pooled far-site SD σ_far,d (meV)
30
25
20
15
10
5
0
Sr
Ca
Cl
Cs
GA
c
Elements gate G1
Cl (passes)
Ca (fails G1)
outside bound
|Δ| (meV, log scale)
100
10
1
+49.4
≤ 5
≤ 3
−1.9
+1.6
+1.6
|M| ≤ 0.01 μB; valence matches (both)
Cutoff
60 vs 50 Ry
Spin
nspin 2 vs 1
d
A2 layout: highest rate above dashed
dashed: 1.15 × rate(48) = 12.19
Configurations per node-hour
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10.60
13.10
12.59
48 ranks
64 ranks
2 × 32